Chelsio Communications

Accelerate

Classification & Filtering

Traffic Management



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Chelsio T5/T6 Packet Classification and Filtering Features

1. Supported Match Fields with Optional Masks
   - Up to 500K exact-match and 2K wildcard rules, 20M/sec lookup rate, 200K updates/sec.
   - Ethernet - Destination Ethernet Address, Ethertype, Inner VLAN, Outer VLAN.
   - IPv4/IPv6 - Source and Destination IP Address, TOS, Protocol, IPv4 Don't Frag/More Frag.
   - TCP/UDP - Source and Destination Port Address.
   - Match Encapsulated packets such as GENEVE, VXLAN, etc.
   - Match packets coming from specified PF/VF.
2. Supported Actions
   - Drop matched packet in hardware.
   - Redirect packet to specified ingress queue.
   - Redirect packet to specified port in hardware with optional header re-write, including:
     - Source and Destination Ethernet Address swap or rewrite.
     - Insertion, Deletion and Rewrite of VLAN headers.
     - Rewrite IP addresses and TCP/UDP ports to achieve NAT.
3. Supported Statistics
   - Per - Filter hit count showing number of times the filter is hit.
   - Per - Filter byte count showing number of bytes that hit the filter.

Chelsio T5/T6 Traffic Management Features

1. Supported Tx Traffic Rate Limiting Scheduling Classes:
   - Class Based Rate Limiting:
     - Per-Flow: Each individual flow bound to the class is rate-limited to the specified rate.
     - Per-Class: All flows bound to the class are aggregate rate-limited to the specified rate.
     - Absolute and relative (%) rate limiting.
     - Bits Per Second and Packets Per Second rate limiting.
   - Class Based Weighted Round Robin Rate Limiting:
     - Weight based rate limiting in a Round Robin manner.
   - Channel Based Rate Limiting:
     - Per-Channel: All flows going through the specified port (channel) are rate-limited to specified rate.
     - Absolute and relative (%) rate limiting.
     - Bits Per Second and Packets Per Second rate limiting.
2. Supported Scheduling Class Bind Options:
   - Bind a Tx queue to a specified traffic scheduling class.
   - Bind a PF/VF to a specified traffic scheduling class.
   - Bind an individual connection to a specified traffic scheduling class.

Chelsio T5/T6 Offload for TC

Currently supported u32 Offload:

- Match packets based on supported IPv4/IPv6-TCP/UDP fields.
- Actions to drop matched packet and redirect matched packet to specified port in hardware.

```
# ethtool -K enp4s0f4 hw-tc-offload on
#
# ./tc qdisc add dev enp4s0f4 handle ffff: ingress
#
# ./tc filter add dev enp4s0f4 parent ffff: protocol ip handle 1: u32 divisor 1
#
# ./tc filter add dev enp4s0f4 parent ffff: protocol ip \
>   u32 ht 800: order 1 link 1: \
>   offset at 0 mask 0f00 shift 6 plus 0 eat \
>   match ip protocol 17 0xff \
>   match ip src 1.1.1.0/24
#
#
# ./tc filter add dev enp4s0f4 parent ffff: protocol ip \
>   u32 ht 1: order 1 \
>   match udp src 5000 0xffff \
>   match udp dst 5001 0xffff \
>   action drop
#
```

```
#
# cat /sys/kernel/debug/cxgb4/0000\:04\:00.4/filters
[[Legend: '!' => locked; '+' => pending set; '-' => pending clear]]
 Idx      Hits Port  Prot              LIP              FIP       LPORT       FPORT Action
   1    782639  0/7 11/ff  00000000/00000000  01010100/ffffff00 1389/ffff 1388/ffff Drop
#
#
```

Chelsio T5/T6 Offload for TC

- Supported Chelsio Terminator 5 10G/40G Series:
  - T580-OCP-SO
  - T580-LP-CR
  - T580-SO-CR
  - T580-CR
  - T540-CR
  - T520-OCP-SO
  - T520-LL-CR
  - T520-SO-CR
  - T520-CR
  - T520-BT

- Support for next generation Chelsio Terminator 6 10G/25G/40G/50G/100G series will be made available soon.

Chelsio T5/T6 Offload for TC

Future Plan / Outstanding Challenges:

- u32 Offload:
  - Match based on Ethernet header fields, Inner and Outer VLAN fields, Encapsulated Packets (GENEVE, VXLAN, etc.), PF/VF.
  - Action to rewrite headers of matched packets.
  - Fetch statistics of filters hit in hardware.

- Tx Traffic Management Offload:
  - Support to offload appropriate traffic classes to underlying hardware.
    - Absolute vs relative (%) rate limit with respect to underlying link speed.
    - Bits Per Second vs Packets Per Second rate limit.
    - Rate limit based on weights in a round robin manner.
  - Per-Flow rate limiting: If a class is rate limited to 10 Mbps and if there are 5 connections bound to it, each connection gets rate-limited to 10 Mbps, instead of 2 Mbps each.
  - Per-Channel rate limiting: All flows going through the specified port get rate limited.
  - Ability to perform rate limiting on PF/VF.